

DHRUV BHAGTANI

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RESEARCH INTERESTS

I was awarded my Ph.D. in 2025 in the Climate and Fluids Physics Group, Australian National University, Australia. My doctoral research focused on better understanding the drivers of large-scale ocean circulation, with an emphasis on gyres (swirling water masses in the ocean). I started a postdoctoral position at the Princeton University in 2025, where I focus on improving estimates of oceanic heat, oxygen, and carbon budgets using multi-scale ocean modelling and machine learning approaches.

WORK EXPERIENCE

- Postdoctoral Scholar, Department of Geosciences, Princeton University Feb '25 - Present
Supervisor: [Laure Resplandy](#)

EDUCATION

- Australian National University Oct '20 - Dec '24
Ph.D. in Earth Sciences, Supervisory Panel: [Andy Hogg](#), [Navid Constantinou](#) and [Ryan Holmes](#)
- Thesis: "Ocean Gyres Driven by Wind Stress and Surface Buoyancy Forcing"
- Indian Institute of Technology Madras Aug '16 - Jun '20
BTech (Honours) in Naval Architecture and Ocean Engineering, CGPA: 9.19/10 (School Rank: 1)
- Honours thesis: "Parallelised Finite Element-Based Solver for Cyclone Inundation in the East Coast of Tamil Nadu."

PUBLICATIONS

- **D. Bhagtani**, A.M. Hogg, R.M. Holmes, N.C. Constantinou, and Khatri, H., 2025: "Asymmetric Response of the North Atlantic Gyres to the North Atlantic Oscillation", Journal of Geophysical Research: Oceans (doi.org/10.1029/2024JC021997).
- A.J. Barnes, N.C. Constantinou, A.H. Gibson, A.E. Kiss, C. Chapman, J. Reilly, **D. Bhagtani**, and L. Yang, 2024: "regional-mom6: A Python package for automatic generation of regional configurations for the Modular Ocean Model 6", Journal of Open Source Software, (<https://doi.org/10.21105/joss.06857>).
- **D. Bhagtani**, A.M. Hogg, R.M. Holmes, and N.C. Constantinou, 2024 (*in review*): "Unraveling how winds and surface heat fluxes control the Atlantic meridional heat transport", Geophysical Research Letters, (<https://arxiv.org/abs/2401.14230>).
- **D. Bhagtani**, A.M. Hogg, R.M. Holmes, and N.C. Constantinou, 2023: "Surface heating steers planetary-scale ocean circulation", Journal of Physical Oceanography, 53(10), 2375-2391 <https://doi.org/10.1175/JPO-D-23-0016.1>.
- **D. Bhagtani**, "Parallelised Finite Element Based Solver for Cyclone Inundation in the East Coast of Tamil Nadu.", Honours Thesis, Indian Institute of Technology Madras, June 2020 <http://dx.doi.org/10.13140/RG.2.2.33408.05120>.
- **D. Bhagtani** and N. Saha, "Extended Kalman Filtering for Estimating Drag and Inertia Coefficients for Slender Offshore Structures.", *Proceedings of the ASME 2019 38th International Conference on Ocean, Offshore and Arctic Engineering*. Structures, Safety, and Reliability. Scotland, UK. June 9–14, 2019 <https://doi.org/10.1115/OMAE2019-96630>.

SCHOLARSHIPS AND PRIZES

Scholarships

- University Research Fellowship - AU\$ 36,652 per annum *Oct '20 - Dec '24*
- HDR Fee Remission Merit Scholarship - AU\$ 44,470 per annum *Oct '20 - Dec '24*
- HDR Supplementary Scholarship - AU\$ 2,000 per annum *Oct '20 - Dec '24*

Prizes

- Participating in the Physical Oceanography Dissertation Symposium - 5,100 US\$ *Oct '24*
- FDSE Summer School Bursary - 2,000 € *Jun '22*
- American Bureau of Shipping Prize - INR 15,000 *Jun '20*
- Alumni Fund Travel Grant - INR 50,000 *Jun '19*
- Future Research Talent Fellowship - AU\$ 6,000 *Dec '18*

Contributions to Grants

- National Computational Merit Allocation Scheme (Lead CI: Dr. Adele Morrison; Won compute resources worth: (i) 1,920,000 AU\$ in 2023, and (ii) 1,784,000 AU\$ in 2024): Contributed to writing proposals and progress reports. Among other papers, Bhagtani et al. (2023), Bhagtani et al. (2024a; in review), and Bhagtani et al. (2024b; in review) are scientific outputs of this grant.

MEDIA APPEARANCES AND OUTREACH

- Interview by the Australian Research Council Centre of Excellence for Climate Extremes during a 3-day hackathon (12-14 Mar 2024) to evaluate Australia's climate model, ACCESS and provide scientific input for the Coupled Model InterComparison Project 7.
- Co-authored the Australian Research Centre of Excellence for Climate Extremes "State of Weather and Climate Extremes Report 2023", Mar 2024. Also appeared at 7News, 9News, and UNSW Sydney news.
- Interview by the College of Science, Australian National University in collaboration with the National Computational Infrastructure on being awarded the Future Research Talent Award to conduct research on the North Atlantic Ocean, Mar 2023.

COMMUNITY SERVICE

- Reviewer for Climate Dynamics, Journal of Climate, Nature Communications, Journal of Physical Oceanography, and Journal of Geophysical Research: Oceans.
- Scientific steering committee, Ocean Modelling & Observations Workshop, Canberra, Australia, Jul 2024.
- Volunteer for annual field trips to Mount Kosciuszko to demonstrate rock formations and climate change to first year students at the ANU (2022-2024).
- Notetaker for the CMIP7 Hackathon, Aspendale, Australia, Mar 2024.
- Session chair for COSIMA workshop, Canberra, Australia, Sep 2023.
- Student Worker for Ocean Sciences Meeting, Online, Feb 2022.
- Led weekly GFD discussions in the Climate and Fluid Physics Group, Australian National University, 2022.

TALKS

- **Bhagtani D.**, Hogg A.M., Holmes R.M., and Constantinou N.C., "The response of oceanic meridional heat transport to varying surface forcing", COSIMA Meeting, Canberra, Australia, Mar 2024.
- **Bhagtani D.**, Hogg A.M., Holmes R.M., and Constantinou N.C., "Unravelling how winds and surface heat fluxes control the Atlantic Ocean's meridional heat transport", *Department of Atmospheric and Oceanic Sciences*, University of California Los Angeles, USA, Feb 2024.
- **Bhagtani D.**, Hogg A.M., Holmes R.M., and Constantinou N.C., "The response of oceanic meridional heat transport to varying surface forcing", Australian Meteorological and Oceanographic Society Conference, Canberra, Feb 2024.

- **Bhagtani D.**, Hogg A.M., Constantinou N.C., and Holmes R.M., “How does buoyancy forcing affect ocean gyres?”, COSIMA Meeting, Canberra, Australia, Oct 2022.
- **Bhagtani D.**, Hogg A.M., Constantinou N.C., and Holmes R.M., “The interplay between winds and surface buoyancy in driving ocean gyres.”, Gerringong retreat, May 2022.
- **Bhagtani D.**, Hogg A.M., Constantinou N., and Holmes R.M., “How does buoyancy forcing influence ocean gyres?”, *Oceans Sciences Institute*, University of Western Australia, Perth, Australia, Apr 2022.
- **Bhagtani D.**, Hogg A.M., Constantinou N., and Holmes R.M., “The interplay between winds and surface buoyancy in driving ocean gyres.”, Ocean Sciences Meeting, Virtual, Feb 2022.
- **Bhagtani D.** and Ranganathan T., “PhD vs PostDoc: Career Opportunities Ahead”, *International Alumni Relations*, Indian Institute of Technology Madras, Virtual, Jun 2021.
- **Bhagtani D.**, “Effects of Surface Waves on the Marginal Ice Zone”, Technical Seminar, *Department of Ocean Engineering*, Indian Institute of Technology Madras, Virtual, Jun 2020.

POSTERS

- **Bhagtani D.** and Resplandy L., “Past and future ocean deoxygenation”, InMOS Annual Meeting, Santa Barbara, USA, Sep 2025.
- **Bhagtani D.**, Hogg A.M., Holmes R.M., and Constantinou N.C., “The response of Atlantic meridional heat transport to varying surface forcing”, Ocean Sciences Meeting, New Orleans, USA, Feb 2024.
- Constantinou N.C., Auger M., Beucher M., **Bhagtani D.**, and others, “A community-driven cookbook of recipes for analysing global ocean-sea ice model output”, Ocean Sciences Meeting, New Orleans, USA, Feb 2024.
- **Bhagtani D.**, Hogg A.M., Holmes R.M., and Constantinou N.C., “The interplay between wind stress and surface buoyancy forcing in driving ocean circulation”, International Union of Geodesy and Geophysics, Berlin, Germany, Jul 2023.
- **Bhagtani D.**, Hogg A.M., Constantinou N.C. and Holmes R.M., “The interplay between winds and surface buoyancy in driving ocean gyres”, Atmospheric and Oceanic Fluid Dynamics Conference, *American Meteorological Society*, Breckenridge, USA, Jun 2022.

WORKSHOPS, SUMMER/WINTER SCHOOLS, AND HACKATHONS

- **Bhagtani D.**, Hogg A.M., Holmes R.M., and Constantinou N.C., “Unraveling how winds and surface heat fluxes control the Atlantic Ocean’s meridional heat transport”, CLEX Workshop, Hunter Valley, Australia, Nov, 2024.
- **Bhagtani D.**, Hogg A.M., Holmes R.M., and Constantinou N.C., “What drives ocean gyres?”, *Physical Oceanography Dissertation Symposium*, Kauai, Hawaii, Oct 2024. Featured in [national news](#).
- **Bhagtani D.**, Hogg A.M., Holmes R.M., and Constantinou N.C., “The response of Atlantic meridional heat transport to varying surface forcing”, Ocean Modeling and Observations Workshop, Canberra, Australia, Jul 2024.
- CMIP7 ACCESS Evaluation Hackathon, CSIRO, Aspendale, Australia, Mar 2024.
- **Bhagtani D.**, Hogg A.M., Holmes R.M., and Constantinou N.C., “Unravelling how winds and surface buoyancy forcing control the Atlantic meridional heat transport”, *CLEX Workshop*, Nov 2023. ([Link](#))
- **Bhagtani D.**, Hogg A.M., Holmes R.M., and Constantinou N.C., “The interplay between wind stress and surface buoyancy forcing in driving the large-scale ocean circulation”, *COSIMA Workshop*, Canberra, Australia, Sep 2023.
- Physics of the Ocean Summer school, Bad Honnef Physics School, Bonn, Germany, Jul 2023.
- ARC Center of Excellence for Climate Extremes Winter School (Theme: Observations in the Climate System), Institute for Marine and Antarctic Studies, Hobart, Australia, Jun 2023.
- **Bhagtani D.**, Hogg A.M., Constantinou N.C. and Holmes R.M., “The interplay between wind stress and surface buoyancy fluxes in driving ocean gyres.”, *CLEX Workshop*, Lorne, Victoria, Australia, Nov 2022.
- **Bhagtani D.**, Hogg A.M., Constantinou N.C. and Holmes R.M., “The interplay between wind stress and surface buoyancy fluxes in driving ocean gyres.”, *COSIMA Workshop*, Tasmania, Australia, Nov 2022. ([Link](#))
- Scientific Paper Writing Workshop, ARC Center of Climate Extremes, Presenter: Dr. Peter Rothlisberg from CSIRO, Oct 2022.

- Fluid Dynamics and Sustainability of the Environment (FDSE) Summer School, *Ecole Polytechnique*, Jun-Jul 2022.
- ARC Center of Excellence for Climate Extremes Winter School (Theme: Atmosphere and Ocean Dynamics), Australian National University, Canberra, Australia, Jun 2022.

SELECT CODE CONTRIBUTIONS

- COSIMA-recipes, e.g., created a beginner-friendly notebook to estimate meridional heat transport in ACCESS-OM2. ([Link](#))
- Improvements in the Regional MOM6 example code, COSIMA. ([Link](#))
- Oceananigans, CliMA, e.g., rectified a bug in the shallow water code. ([Link](#))
- “CMIP6 Analysis Ready Data workflow: turning big climate projection data into useful inputs for modelling or analysis”, *Ocean Hack Week*, Virtual, Aug 2021. ([Link](#))

TEACHING AND DEMONSTRATING ACTIVITIES

- Student Ambassador for Future Research Talent Internship Programme, ANU *Mar '22 - Dec '24*
 - Providing information and advice on the student experience to prospective students, assisting the International Relations and Partnerships team in event coordination and administration.
 - Presented my student experience at ANU as an FRT scholar to students at various events.
 - Organised annual international panel discussions with doctoral and post-doctoral fellows.
- Course Demonstrator at the Research School of Earth Sciences, ANU
 - Climate Dynamics: 3rd year course on understanding weather and climate. *Feb '24 - Jun '24*
 - Climate Science: 2nd year course on climate change. *Jul '22 - Nov '22; Jul '23 - Nov '23*
 - Fluid Physics and Plasma: 3rd year course on the physics of the ocean. *Jul '22 - Nov '22*
 - The Blue Planet: 1st year course on earth and the climate dynamics. *Feb '22 - Jun '22; Feb '23 - Jun '23*

COMPUTING SKILLS

- Programming Languages: FORTRAN, C++, Python, Julia
- Programming Environments: MATLAB, GNU Octave, CUDA, OpenMP, GitHub
- Operating Systems: MacOS, Windows, Linux

INDUSTRIAL INTERNSHIPS

- Internship at Detect Technologies, Chennai, India *Mar '18 - Jul '18*
Guide: Prof. Krishnan Balasubramanian, Department of Mechanical Engineering, IIT Madras
 - Implemented Kalman Filtering in flight data to improve live tracking of drones (inside pipes for inspection) with a precision of $\sim 85\%$.
- Shipyard Training at Cochin Shipyard Limited (CSL), Kerala, India *Jun '18*
 - Completed a three week training in three departments - Ship Design, Ship Building and Ship Repair.